

MA1521 L1 Week 02 Wed 20260819

For what value(s) of c will $\frac{1}{5}\mathbf{i} - \frac{1}{5}\mathbf{j} + c\mathbf{k}$ be a unit vector?

$$\left(\frac{1}{5}, -\frac{1}{5}, c\right) = \begin{pmatrix} 1/5 \\ -1/5 \\ c \end{pmatrix}$$

$$(\text{Length of vector})^2 = \begin{pmatrix} 1/5 \\ -1/5 \\ c \end{pmatrix} \cdot \begin{pmatrix} 1/5 \\ -1/5 \\ c \end{pmatrix} = \frac{2}{25} + c^2. \quad \text{For this vector to be unit vector,}$$

must have $\frac{2}{25} + c^2 = 1$

i.e. $\boxed{c = \pm \frac{\sqrt{23}}{5}}$

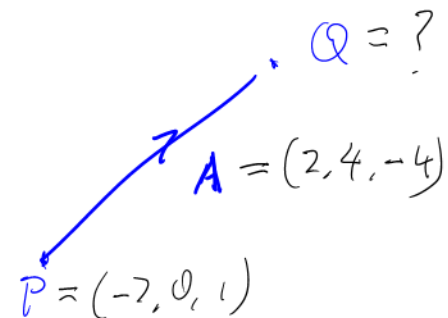
A vector $\mathbf{A}$ has magnitude 6 and direction $(\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})/3$.

If its tail is at $P = (-2, 0, 1)$, where is its head? $Q = ?$

$$\mathbf{A} = 6 \cdot (\mathbf{i} + 2\mathbf{j} - 2\mathbf{k})/3 = 2\mathbf{i} + 4\mathbf{j} - 4\mathbf{k}$$

$$\text{so head is } Q = (-2, 0, 1) + (2, 4, -4)$$

$$= \underline{(0, 4, -3)}$$

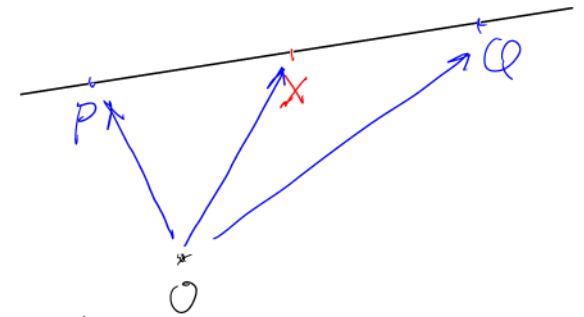


- a) Let P and Q be two points in space, and X the midpoint of the line segment PQ .
Let O be an arbitrary fixed point; show that as vectors, $OX = \frac{1}{2}(OP + OQ)$.
- b) With the notation of part (a), assume that X divides the line segment PQ in the ratio $r : s$, [where $r + s = 1$]. Derive an expression for OX in terms of OP and OQ .

a) Saying that X the midpoint of the line segment PQ .

means $\vec{PX} = \vec{XQ}$ (and both equal to $\frac{1}{2}\vec{PQ}$)

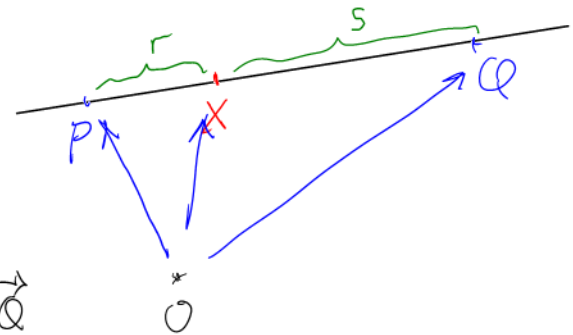
$$\begin{aligned} \text{so } \vec{OX} &= \vec{OP} + \vec{PX} = \vec{OP} + \frac{1}{2}\vec{PQ} \\ &= \vec{OP} + \frac{1}{2}(\vec{OQ} - \vec{OP}) = \frac{1}{2}(\vec{OP} + \vec{OQ}) \end{aligned}$$



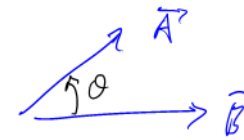
b) Saying that X divides the line segment PQ in the ratio $r : s$.

means $\vec{PX} = \frac{r}{s} \vec{XQ}$

$$\left(\begin{aligned} \vec{PX} + \vec{XQ} &= \vec{PQ} \\ \frac{r}{s} \vec{XQ} + \vec{XQ} &= \vec{PQ} \\ \frac{r+s}{s} \vec{XQ} &= \vec{PQ} \end{aligned} \right) \Rightarrow \begin{aligned} \vec{PX} &= \vec{PQ} - \frac{s}{r+s} \vec{PQ} \\ &= \frac{r}{r+s} \vec{PQ} \end{aligned}$$



$$\text{so } \vec{OX} = \vec{OP} + \vec{PX} = \vec{OP} + \frac{r}{r+s} \vec{PQ} = \vec{OP} + \frac{r}{r+s} (\vec{OQ} - \vec{OP}) = \frac{s}{r+s} \vec{OP} + \frac{r}{r+s} \vec{OQ}$$



$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{|\vec{A}| \cdot |\vec{B}|}$$

Find the angle between the vectors

a) $\mathbf{i} - \mathbf{k}$ and $4\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}$

$$\cos \theta = \frac{\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \cdot \begin{pmatrix} 4 \\ 4 \\ -2 \end{pmatrix}}{|\begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}| \cdot |\begin{pmatrix} 4 \\ 4 \\ -2 \end{pmatrix}|} = \frac{6}{\sqrt{2} \sqrt{36}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \theta = \frac{\pi}{4}$$

b) $\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $2\mathbf{i} - \mathbf{j} + \mathbf{k}$.

$$\cos \theta = \frac{\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}}{|\begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}| \cdot |\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}|} = \frac{3}{\sqrt{6} \cdot \sqrt{6}} = \frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{3}$$

Tell for what values of c the vectors $(c\mathbf{i} + 2\mathbf{j} - \mathbf{k})$ and $(\mathbf{i} - \mathbf{j} + 2\mathbf{k})$ will

a) be orthogonal $\theta = \frac{\pi}{2}$ b) form an acute angle $0 < \theta < \frac{\pi}{2}$

$$\cos \theta = \frac{A(c) \cdot B}{|A(c)| \cdot |B|} = \frac{c - 2 - 2}{\sqrt{c^2 + 5} \sqrt{6}}$$

so a) orthog iff $\cos \theta = 0$ iff $c = 4$

b) acute iff $\cos \theta > 0$ iff $c > 4$

Calculate the value of the determinants

a) $\begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix}$ b) $\begin{vmatrix} 3 & -4 \\ -1 & -2 \end{vmatrix}$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$$

$$\begin{aligned} &= (1)(-1) - (4)(2) &= (3)(-2) - (-4)(-1) \\ &= -1 - 8 = \underline{\underline{-9}} &= -6 - 4 = \underline{\underline{-10}} \end{aligned}$$

Calculate $\begin{vmatrix} -1 & 0 & 4 \\ 1 & 2 & 2 \\ 3 & -2 & -1 \end{vmatrix}$ using the Laplace expansion by the cofactors of:

a) the first row

b) the first column

a) $+(-1) \cdot \begin{vmatrix} 0 & 4 \\ 2 & 2 \\ -2 & -1 \end{vmatrix} - (0) \begin{vmatrix} -1 & 4 \\ 1 & 2 \\ 3 & -1 \end{vmatrix} + (4) \begin{vmatrix} -1 & 0 \\ 1 & 2 \\ 3 & -2 \end{vmatrix}$

$$= (-1)(-2+4) + 4(-2-6) = \underline{\underline{-34}}$$

b) $+(-1) \begin{vmatrix} 0 & 4 \\ 2 & 2 \\ -2 & -1 \end{vmatrix} - (1) \begin{vmatrix} -1 & 4 \\ 1 & 2 \\ 3 & -1 \end{vmatrix} + (3) \begin{vmatrix} -1 & 0 \\ 1 & 2 \\ 3 & -2 \end{vmatrix}$

$$= (-1)(-2+4) - (+8) + (3)(-8) = \underline{\underline{-34}}$$

Find $\mathbf{A} \times \mathbf{B}$ if

a) $\mathbf{A} = \underline{\mathbf{i}} - \underline{2\mathbf{j}} + \underline{\mathbf{k}}, \quad \mathbf{B} = \underline{2\mathbf{i}} - \underline{\mathbf{j}} - \underline{\mathbf{k}}$ b) $\mathbf{A} = \underline{2\mathbf{i}} - \underline{3\mathbf{k}}, \quad \mathbf{B} = \underline{\mathbf{i}} + \underline{\mathbf{j}} - \underline{\mathbf{k}}.$

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & -2 & 1 \\ 2 & -1 & -1 \end{vmatrix} \begin{matrix} \leftarrow \mathbf{A} \\ \leftarrow \mathbf{B} \end{matrix}$$

$$= \mathbf{i} \begin{vmatrix} -2 & 1 \\ -1 & -1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 1 & 1 \\ 2 & -1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 1 & -2 \\ 2 & -1 \end{vmatrix}$$

$$= 3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k}$$

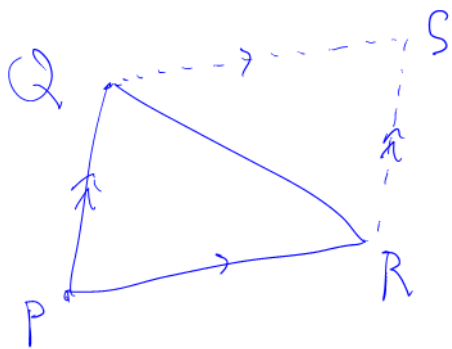
$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 0 & -3 \\ 1 & 1 & -1 \end{vmatrix} \begin{matrix} \leftarrow \mathbf{A} \\ \leftarrow \mathbf{B} \end{matrix}$$

$$= \mathbf{i} \begin{vmatrix} 0 & -3 \\ 1 & -1 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 2 & -3 \\ 1 & -1 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix}$$

$$= 3\mathbf{i} - \mathbf{j} + 2\mathbf{k}$$

Find the area of the triangle in space having its vertices at the points

$P : (2, 0, 1), \quad Q : (3, 1, 0), \quad R : (-1, 1, -1).$



$\text{Area } \triangle PQR = \frac{1}{2} \text{ Area // } \text{of } \text{parallelogram } PQRS$

$$= \frac{1}{2} | \vec{PR} \times \vec{PQ} |$$

$$= \frac{1}{2} \left| \begin{pmatrix} (-1, 1, -1) \\ -(2, 0, 1) \end{pmatrix} \times \begin{pmatrix} (3, 1, 0) \\ -(2, 0, 1) \end{pmatrix} \right|$$

$$= \frac{1}{2} \text{ length of } \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 1 & -2 \\ 1 & 1 & -1 \end{vmatrix} = \frac{1}{2} | (1, -5, -4) |$$

$$= \frac{\sqrt{42}}{2}$$

$$ax + by + cz = d, \quad a, b, c, d \in \mathbb{R}.$$

Find the equations of the following planes:

- through $(2, 0, -1)$ and perpendicular to $\mathbf{i} + 2\mathbf{j} - 2\mathbf{k}$
- through the origin, $(1, 1, 0)$, and $(2, -1, 3)$
- through $(1, 0, 1)$, $(2, -1, 2)$, $(-1, 3, 2)$
- through the points on the x , y and z -axes where $x = a$, $y = b$, $z = c$ respectively
- through $(1, 0, 1)$ and $(0, 1, 1)$ and parallel to $\mathbf{i} - \mathbf{j} + 2\mathbf{k}$

a)

$$\left(\underbrace{(x, y, z) - (2, 0, -1)}_{\text{blue vector}} \right) \cdot \underline{\mathbf{i}} + \underline{2\mathbf{j}} - \underline{2\mathbf{k}} = 0$$

$$(x-2) \cdot 1 + (y-0) \cdot 2 + (z+1) \cdot (-2) = 0$$

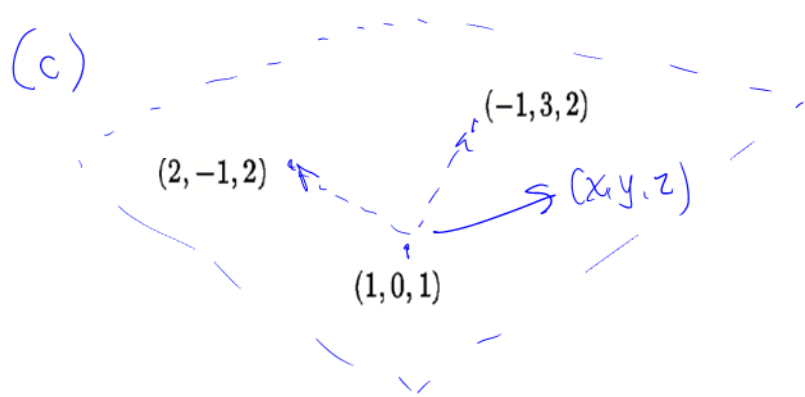
$$\text{i.e. } \underline{x + 2y - 2z = 4}$$

b)

$$\left(\underbrace{(x, y, z)}_{\text{blue vector}} \right) \cdot \left(\underbrace{(1, 1, 0) \times (2, -1, 3)}_{(3, -3, -3)} \right) = 0$$

$$\text{i.e. } (x, y, z) \cdot \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 0 \\ 2 & -1 & 3 \end{vmatrix} = 0$$

$$\text{i.e. } 3x - 3y - 3z = 0 \quad \text{i.e. } \underline{x - y - z = 0}$$



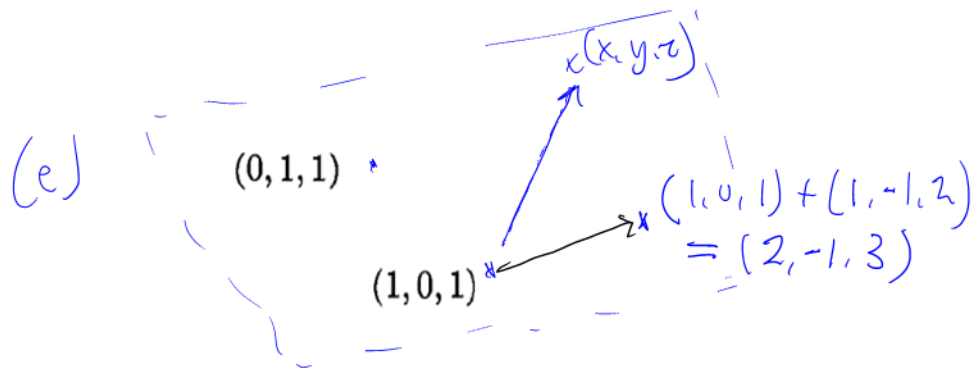
$$\bullet \left((x,y,z) - (1,0,1) \right) \perp \begin{pmatrix} (2,-1,2) - (1,0,1) \\ (-1,3,2) - (1,0,1) \end{pmatrix}$$

$$\text{i.e. } (x-1, y, z-1) \cdot \begin{pmatrix} i & j & k \\ 1 & -1 & 1 \\ -2 & 3 & 1 \end{pmatrix} = 0$$

$$(-4, -3, 1)$$

$$\text{i.e. } (x-1)(-4) + y(-3) + (z-1) = 0$$

$$\text{i.e. } -4x - 3y + z = -3$$



$$\bullet \text{ plane is } \parallel \text{ to } \mathbf{i} - \mathbf{j} + 2\mathbf{k} \text{ (given)}$$

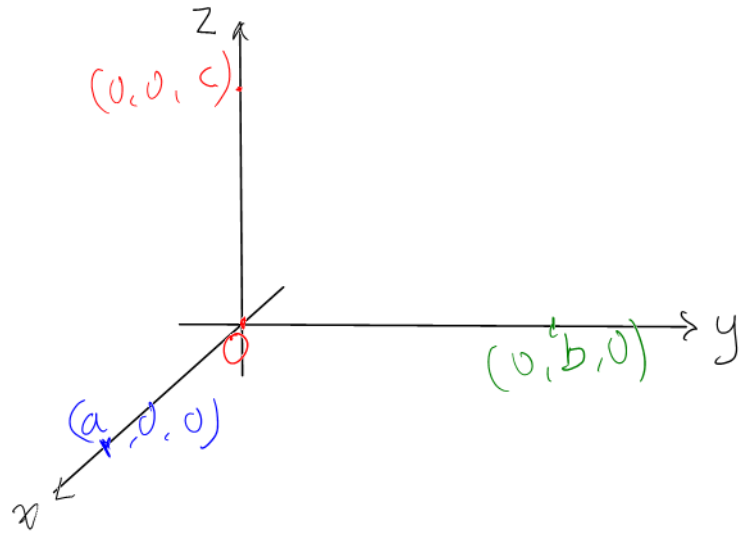
$$\text{and also to } (0,1,1) - (1,0,1) = -\mathbf{i} + \mathbf{j}$$

$$\bullet \text{ plane is } \perp \text{ to } (\mathbf{i} - \mathbf{j} + 2\mathbf{k}) \times (-\mathbf{i} + \mathbf{j})$$

$$\text{i.e. } ((x,y,z) - (1,0,1)) \cdot \begin{pmatrix} i & j & k \\ 1 & -1 & 2 \\ -1 & 1 & 0 \end{pmatrix} = 0$$

$$\text{i.e. } x + y = 1$$

d) through the points on the x , y and z -axes where $x = a$, $y = b$, $z = c$ respectively



eqn: $Px + Qy + Rz = S$, $P, Q, R, S \in \mathbb{R}$.

• $Pa = S$

• $Qb = S$

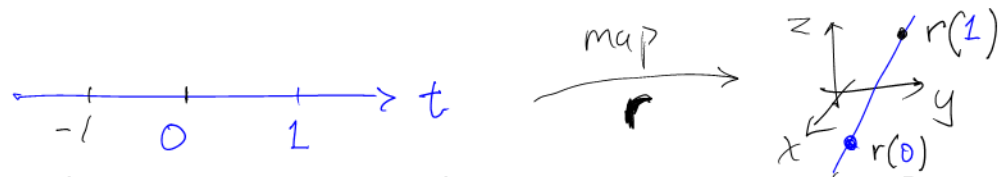
• $Rc = S$

• Can set $S = 1$ to get $P = \frac{1}{a}$

$$Q = \frac{1}{b}$$

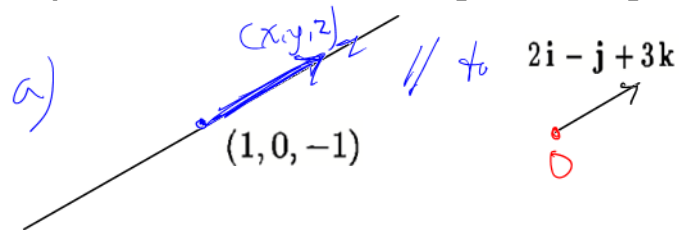
$$R = \frac{1}{c}$$

so eqn is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$



Find in parametric form the equations for

- the line through $(1, 0, -1)$ and parallel to $2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$
- the line through $(2, -1, -1)$ and perpendicular to the plane $x - y + 2z = 3$
- all lines passing through $(1, 1, 1)$ and lying in the plane $x + 2y - z = 2$

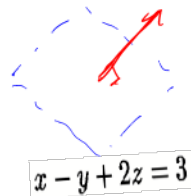
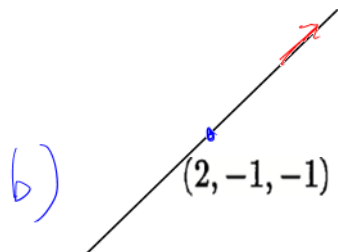


$$((x, y, z) - (1, 0, -1)) \parallel 2\mathbf{i} - \mathbf{j} + 3\mathbf{k}$$

$$\text{i.e. } (x-1, y, z+1) = t(2\mathbf{i} - \mathbf{j} + 3\mathbf{k})$$

$$\text{i.e. } (x, y, z) = (1+2t, -t, -1+3t)$$

$$\text{i.e. } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$



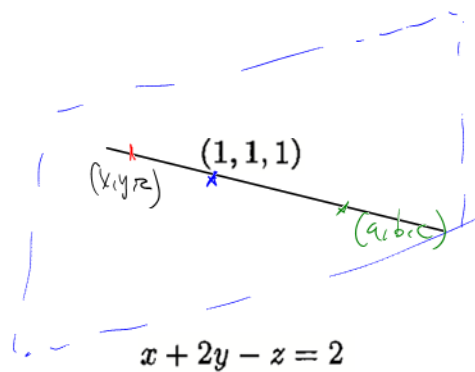
$\Leftrightarrow$ a vector $\perp$ plane is $(1, -1, 2)$

i.e. a vector $\parallel$ line is $(1, -1, 2)$

eqn of line is $(x, y, z) - (2, -1, -1) = t(1, -1, 2)$

$$\text{i.e. } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} + t \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

c)



z plane has eqn $x + 2y - z = 2$

i.e. is h $(1, 2, -1)$

$$a + 2b - c = 2$$

* choose a point (a, b, c) on plane
 $(1, 1, 1)$

Alt Mthd: • choose a vector
 (p, q, r) parallel
to the plane

i.e. (p, q, r) h to $(1, 2, -1)$

$$\text{i.e. } p + 2q - r = 0$$

eqn of line thru $(1, 1, 1)$ and
in direction of (p, q, r)

$$\text{is } (x, y, z) - (1, 1, 1) = t \cdot (p, q, r)$$

$$\text{i.e. } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

where $p + 2q - r = 0$

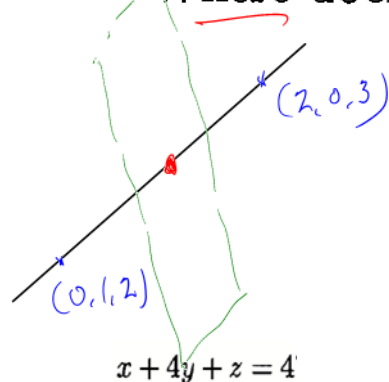
• eqn of line thru $(1, 1, 1)$ and (a, b, c)

$$\text{is } (x, y, z) - (1, 1, 1) = t \cdot ((a, b, c) - (1, 1, 1)) \quad \checkmark$$

$$\text{X i.e. } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -t \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \text{such that} \\ a + 2b - c = 2$$

$$\checkmark \text{ i.e. } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = (1-t) \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \text{such that} \\ a + 2b - c = 2 \\ = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + t \begin{pmatrix} a-1 \\ b-1 \\ c-1 \end{pmatrix} \quad \checkmark$$

Where does the line through $P = (0, 1, 2)$ and $Q = (2, 0, 3)$ intersect the plane $x + 4y + z = 4$?



Line thru P, Q has eqn $\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = t \cdot \left(\begin{pmatrix} 2 \\ 0 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} \right)$

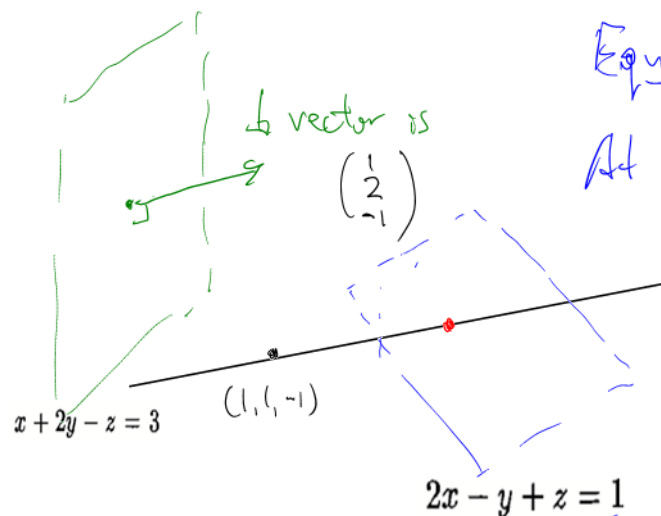
$$\text{i.e. } \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$$

At pt of intersection, $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ lies on plane, i.e. $x + 4y + z = 4$

i.e. coords are $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ -1 \\ 4 \end{pmatrix}$ i.e. $(2t) + 4(1-t) + (2+t) = 4$
i.e. $t = 2$

The line passing through $(1, 1, -1)$ and perpendicular to the plane $x + 2y - z = 3$ intersects the plane $2x - y + z = 1$ at what point?

i.e. line is $\parallel$ to $\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$



Eqn of line is $\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} = t \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$

At pt of intersection, must have $2x - y + z = 1$

$$\text{i.e. } 2(1+t) - (1+2t) + (-1-t) = 1$$

$$\text{i.e. } t = -1$$

i.e. coords are $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}$

